

TechNova - Employee Attrition

End-to-End ML Workflow

Name - MD Sakib
Roll - B24BS2200

Summarize this data																									
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V				
EmployeeID	Age	Gender	Department	JobRole	MonthlyIncome	YearsAtCompan	YearsSinceLastF	JobSatisfaction	WorkLifeBalance	EducationLevel	TrainingHoursLa	OverTime	PerformanceRat	Attrition											
1001	50	Male	IT	System Admin	31288	12	6	2	1	1	10	No	2	No											
1002	36	Male	Sales	Account Manage	87692	13	0	1	2	3	48	No	3	No											
1003	29	Female	Finance	Financial Analys	100436	3	2	2	1	1	20	No	2	Yes											
1004	42	Female	R&D	Lab Technician	61403	7	0	2	1	3	65	No	3	Yes											
1005	40	Female	Sales	Account Manage	106170	4	1	3	1	5	62	No	2	No											
1006	44	Female	R&D	Research Scient	24168	0	0	2	4	3	27	Yes	3	No											
1007	32	Male	HR	HR Generalist	64879	3	2	3	2	5	16	Yes	4	No											
1008	32	Male	IT	System Admin	74871	2	0	4	1	5	5	Yes	5	No											
1009	45	Female	HR	Recruiter	47174	8	8	1	2	3	18	Yes	3	Yes											
1010	57	Female	Sales	Account Manage	125155	7	3	2	4	2	55	No	3	No											
1011	45	Male	Finance	Financial Analys	86853	0	0	1	3	4	27	No	4	Yes											
1012	24	Female	R&D	Research Scient	79046	2	1	2	1	1	65	No	3	Yes											
1013	43	Male	IT	Software Engine	64678	5	4	1	2	4	37	Yes	1	Yes											
1014	23	Male	R&D	Research Scient	41649	0	0	2	2	1	36	Yes	4	No											
1015	45	Male	Sales	Account Manage	142795	6	0	2	4	1	67	No	1	No											
1016	51	Male	R&D	Research Scient	103668	3	0	2	1	5	48	No	4	No											
1017	23	Male	IT	System Admin	104906	0	0	2	3	1	42	Yes	3	No											
1018	42	Male	Sales	Sales Executive	111019	16	5	2	2	1	35	No	1	No											
1019	54	Male	R&D	Research Scient	93002	1	1	2	1	3	57	No	3	No											
1020	33	Female	Finance	Accountant	43803	11	3	2	3	4	12	No	4	No											
1021	43	Female	Sales	Sales Executive	45795	4	2	2	2	3	51	No	3	Yes											
1022	46	Female	Sales	Account Manage	23770	11	5	2	2	1	57	Yes	1	Yes											
1023	48	Female	Finance	Financial Analys	26093	20	10	1	3	5	31	Yes	1	No											
1024	49	Male	HR	HR Generalist	146265	7	6	3	3	3	80	No	3	No											
1025	37	Male	IT	Data Scientist	115311	5	0	2	4	3	59	No	2	No											
1026	36	Female	HR	HR Generalist	32569	7	1	3	1	2	57	No	4	No											
1027	24	Male	R&D	Lab Technician	26190	1	1	2	1	1	55	No	4	No											
1028	58	Female	Finance	Accountant	77824	2	0	2	4	5	27	Yes	3	No											
1029	28	Male	HR	HR Generalist	57070	5	0	1	2	3	0	No	2	Yes											
1030	42	Female	Sales	Account Manage	55197	11	3	4	3	3	41	Yes	1	No											
1031	30	Male	R&D	Lab Technician	17261	2	0	4	3	5	47	No	5	No											
1032	39	Female	R&D	Lab Technician	101795	5	1	1	3	1	39	No	5	No											
1033	25	Male	HR	Recruiter	43933	1	1	3	2	2	62	No	3	No											
1034	46	Female	HR	Recruiter	55583	11	2	4	3	3	75	No	2	No											
1035	35	Male	Finance	Accountant	127082	1	0	1	3	2	71	No	2	No											
1036	30	Female	IT	Data Scientist	100371	5	2	1	4	4	79	No	4	No											
1037	47	Female	HR	HR Generalist	126032	16	0	4	2	1	3	No	1	No											
1038	23	Male	HR	HR Generalist	57604	0	0	3	1	4	37	Yes	2	No											
1039	41	Female	Sales	Account Manage	110074	1	1	1	4	4	26	Yes	5	No											
1040	49	Female	R&D	Research Scient	88869	19	5	2	4	4	8	Yes	1	Yes											
1041	28	Male	IT	Data Scientist	20410	5	4	3	1	1	15	Yes	1	Yes											
1042	29	Female	Sales	Sales Executive	98375	3	1	3	4	1	55	No	3	No											
1043	56	Female	Finance	Accountant	111055	3	0	4	2	5	75	No	2	No											
1044	35	Female	Finance	Accountant	101716	4	2	4	3	2	30	No	3	No											
1045	38	Male	HR	Recruiter	69245	12	2	2	2	2	13	Yes	2	Yes											
1046	57	Male	HR	Recruiter	58056	14	4	1	1	5	24	No	4	No											
1047	25	Female	Finance	Financial Analys	73557	0	0	4	1	5	33	No	3	No											
1048	23	Female	HR	Recruiter	32543	0	0	4	4	4	66	No	3	No											
1049	27	Male	IT	Data Scientist	111038	1	0	3	2	5	23	No	5	No											
1050	25	Female	HR	HR Generalist	18828	1	1	4	2	3	72	No	3	No											
1051	50	Male	R&D	Lab Technician	131656	15	2	4	1	2	25	Yes	2	No											

Convert to table



ARFF Viewer screenshot

KNIME Analytics Platform

Home KNIME_project +

Local - KNIME_project

Nodes > Results

Search: arff

IO Write Read

ARFF Reader ARFF Writer

ARFF Reader

This node dialog is not supported here.

Open dialog

1: Data from ARFF

Rows: 500 | Columns: 15

#	RowID	EmployeeID (in: Number (Float))	Age (in: Number (Float))	Gender (in: String)	Department (in: String)	JobRole (in: String)	MonthlyIncome (in: Number (Float))	YearsAtCompa... (in: Number (Float))	YearsSinceLas... (in: Number (Float))	JobSatisfaction (in: Number (Float))	WorkLifeBalan... (in: Number (Float))	EducationLevel (in: Number (Float))	TrainingHours... (in: Number (Float))	OverTime (in: String)	PerformanceR... (in: Number (Float))	Attrition (in: String)
1	Row0	1,001	50	Male	IT	System Admin	21,288	12	6	2	1	1	10	No	2	No
2	Row1	1,002	36	Male	Sales	Account Manager	87,692	13	0	1	2	3	48	No	3	No
3	Row2	1,003	29	Female	Finance	Financial Analyst	100,435	3	2	2	1	1	20	No	2	Yes
4	Row3	1,004	42	Female	R&D	Lab Technician	61,403	7	0	2	1	3	65	No	3	Yes
5	Row4	1,005	40	Female	Sales	Account Manager	106,170	4	1	3	1	5	62	No	2	No
6	Row5	1,006	44	Female	R&D	Research Scientist	24,168	0	0	2	4	3	27	Yes	3	No
7	Row6	1,007	32	Male	HR	HR Generalist	64,879	3	2	3	2	5	16	Yes	4	No
8	Row7	1,008	32	Male	IT	System Admin	74,871	2	0	4	1	5	5	Yes	5	No
9	Row8	1,009	45	Female	HR	Recruiter	47,174	8	8	1	2	3	18	Yes	3	Yes
10	Row9	1,010	57	Female	Sales	Account Manager	125,155	7	3	2	4	2	55	No	3	No
11	Row10	1,011	45	Male	Finance	Financial Analyst	86,853	0	0	1	3	4	27	No	4	Yes
12	Row11	1,012	24	Female	R&D	Research Scientist	79,046	2	1	2	1	1	65	No	3	Yes
13	Row12	1,013	43	Male	IT	Software Engineer	64,678	5	4	1	2	4	37	Yes	1	Yes
14	Row13	1,014	23	Male	R&D	Research Scientist	41,649	0	0	2	2	1	36	Yes	4	No
15	Row14	1,015	45	Male	Sales	Account Manager	142,795	6	0	2	4	1	67	No	1	No
16	Row15	1,016	51	Male	R&D	Research Scientist	103,668	3	0	2	1	5	48	No	4	No

Weka preprocessing screenshot

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Open file...

Open URL...

Open DB...

Generate...

Undo

Edit...

Save...

Filter

Choose **None**

Apply

Stop

Current relation

Relation: TechNova_Employee_Attrition

Instances: 500

Attributes: 15

Sum of weights: 500

Attributes

All

None

Invert

Pattern

No.	Name
1	☒ EmployeeID
2	☒ Age
3	☒ Gender
4	☒ Department
5	☒ JobRole
6	☒ MonthlyIncome
7	☒ YearsAtCompany
8	☒ YearsSinceLastPromotion
9	☒ JobSatisfaction
10	☒ WorkLifeBalance
11	☒ EducationLevel
12	☒ TrainingHoursLastYear
13	☒ OverTime
14	☒ PerformanceRating
15	☐ Attrition

Remove

Selected attribute

Name: EmployeeID

Missing: 0 (0%)

Distinct: 500

Type: Numeric

Unique: 500 (100%)

Statistic	Value
Minimum	1001
Maximum	1500
Mean	1250.5
StdDev	144.482

Class: Attrition (Nom)

Visualize All

Log

x 0

Status
OK

Weka baseline model results

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose J48 -C 0.25 -M 2

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds 10

% 66

More options...

(Nom) Attrition

Start

Stop

Result list (right-click for options)

12:42:03 - trees.J48

Classifier output

```

| | | | | Age > 52: No (4.0)
| | | | | WorkLifeBalance > 1: Yes (4.0/1.0)
| | | | | JobRole = Software Engineer
| | | | | YearsSinceLastPromotion <= 1: Yes (2.0)
| | | | | YearsSinceLastPromotion > 1: No (5.0/1.0)
| | | | | JobRole = Sales Executive
| | | | | TrainingHoursLastYear <= 47: No (7.0/1.0)
| | | | | TrainingHoursLastYear > 47: Yes (6.0/1.0)
| | | | | JobRole = Accountant
| | | | | WorkLifeBalance <= 1: No (4.0/1.0)
| | | | | WorkLifeBalance > 1: Yes (2.0)
| | | | | JobRole = Data Scientist
| | | | | Gender = Male
| | | | | Age <= 49: No (2.0)
| | | | | Age > 49: Yes (2.0)
| | | | | Gender = Female: No (3.0)
| | | | | JobSatisfaction > 2: No (53.0/14.0)
| | | | | WorkLifeBalance > 2: No (102.0/20.0)
| | | | | JobSatisfaction > 3: No (125.0/9.0)

Number of Leaves : 34

Size of the tree : 58

Time taken to build model: 0.05 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances 329 65.8 %
Incorrectly Classified Instances 171 34.2 %
Kappa statistic 0.1757
Mean absolute error 0.3859
Root mean squared error 0.5129
Relative absolute error 90.1288 %
Root relative squared error 110.9028 %
Total Number of Instances 500

=== Detailed Accuracy By Class ===

TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
0.394 0.223 0.442 0.394 0.416 0.176 0.568 0.379 Yes
0.777 0.606 0.740 0.777 0.758 0.176 0.568 0.698 No
Weighted Avg. 0.658 0.488 0.648 0.658 0.652 0.176 0.568 0.599

=== Confusion Matrix ===

a b <- classified as
61 94 | a = Yes
77 268 | b = No

```

Status

OK

Log

x 0

Overfitting model results

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose J48 -U-M 1

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds 10

% 66

More options...

(Nom) Attrition

Start

Stop

Result list (right-click for options)

12:48:05 - trees.J48

Classifier output

```
| | Age <= 56: No (9.0)
| | Age > 56: Yes (1.0)
| JobRole = Research Scientist
| | EmployeeID <= 1090: Yes (2.0)
| | EmployeeID > 1090: No (17.0)
| JobRole = HR Generalist
| | YearsSinceLastPromotion <= 6
| | | Age <= 24: Yes (1.0)
| | | Age > 24: No (8.0)
| | YearsSinceLastPromotion > 6: Yes (2.0)
| JobRole = Recruiter: No (18.0)
| JobRole = Software Engineer: No (8.0)
| JobRole = Sales Executive
| | EducationLevel <= 2
| | | Age <= 48: No (2.0)
| | | Age > 48: Yes (2.0)
| | EducationLevel > 2: No (11.0)
| JobRole = Accountant: No (13.0)
| JobRole = Data Scientist: No (2.0)

Number of Leaves : 170

Size of the tree : 273

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances 300 60 %
Incorrectly Classified Instances 200 40 %
Kappa statistic 0.0813
Mean absolute error 0.3995
Root mean squared error 0.6285
Relative absolute error 93.3197 %
Root relative squared error 135.8869 %
Total Number of Instances 500

=== Detailed Accuracy By Class ===

          TP Rate  FP Rate  Precision  Recall  F-Measure  MCC  ROC Area  PRC Area  Class
0.387  0.304  0.364  0.387  0.375  0.081  0.540  0.332  Yes
0.696  0.613  0.716  0.696  0.706  0.081  0.540  0.708  No
Weighted Avg.  0.600  0.517  0.607  0.600  0.603  0.081  0.540  0.591

=== Confusion Matrix ===
  a  b  <-- classified as
60 95 |  a = Yes
105 240 |  b = No
```

Status

OK

Log

x 0

Underfitting model results

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose ZeroR

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds

%

10

66

More options...

(Nom) Attrition

Start

Stop

Result list (right-click for options)

12:48:05 - trees.J48

12:48:48 - trees.J48

12:51:01 - rules.ZeroR

Classifier output

Scheme: weka.classifiers.rules.ZeroR
Relation: TechNova_Employee_Attrition
Instances: 500
Attributes: 15
EmployeeID
Age
Gender
Department
JobRole
MonthlyIncome
YearsAtCompany
YearsSinceLastPromotion
JobSatisfaction
WorkLifeBalance
EducationLevel
TrainingHoursLastYear
OverTime
PerformanceRating
Attrition
Test mode: 10-fold cross-validation
=== Classifier model (full training set) ===
ZeroR predicts class value: No
Time taken to build model: 0 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 345 69 %
Incorrectly Classified Instances 155 31 %
Kappa statistic 0
Mean absolute error 0.4281
Root mean squared error 0.4625
Relative absolute error 100 %
Root relative squared error 100 %
Total Number of Instances 500
=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.000	0.000	?	0.000	?	?	0.488	0.305	Yes
	1.000	1.000	0.690	1.000	0.817	?	0.488	0.685	No
Weighted Avg.	0.690	0.690	?	0.690	?	?	0.488	0.567	

=== Confusion Matrix ===
a b <-- classified as
0 155 | a = Yes
0 345 | b = No

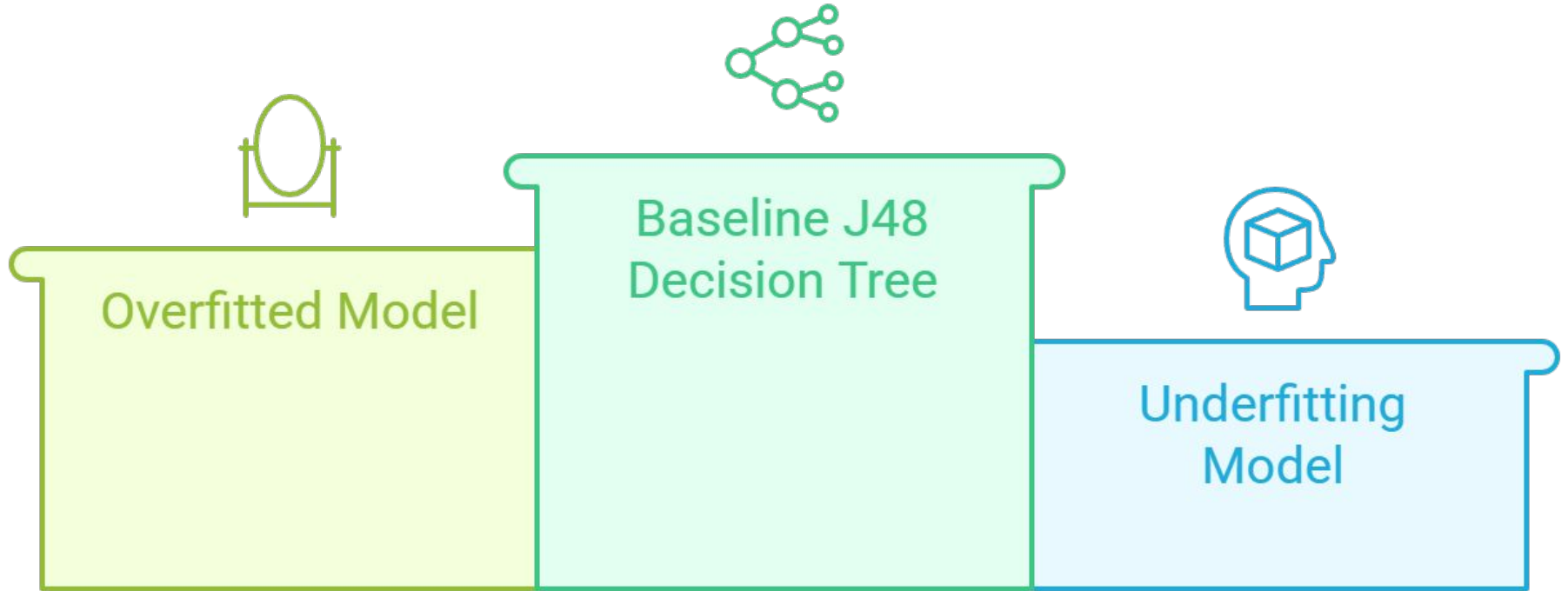
Status

OK

Log

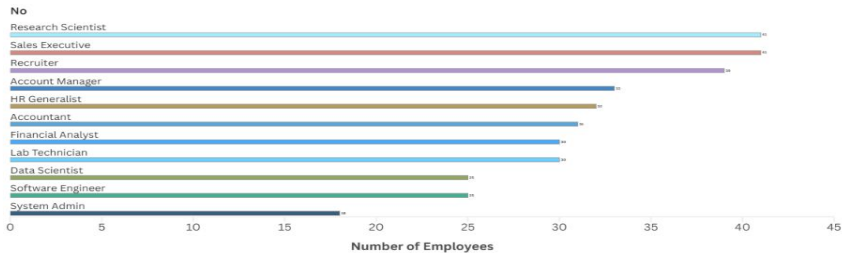
x 0

Model Performance Comparison



Attrition by Department

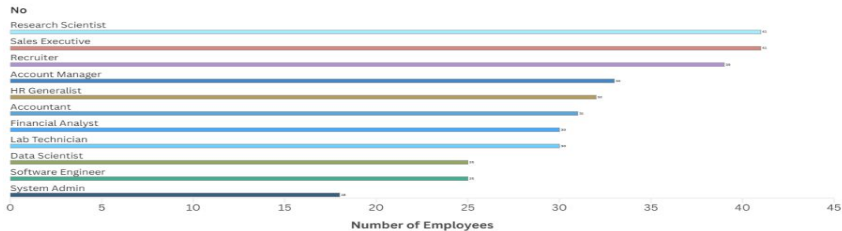
No Yes



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Attrition by Department

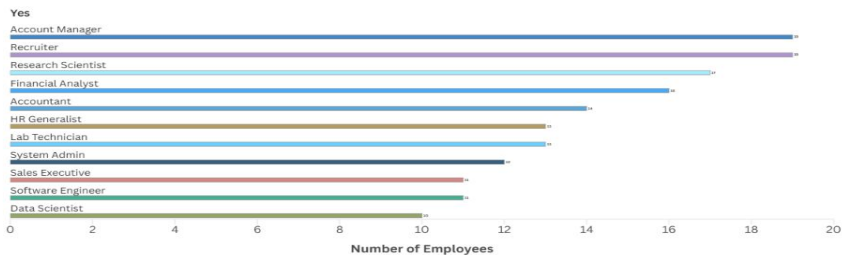
No Yes



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Attrition by Department

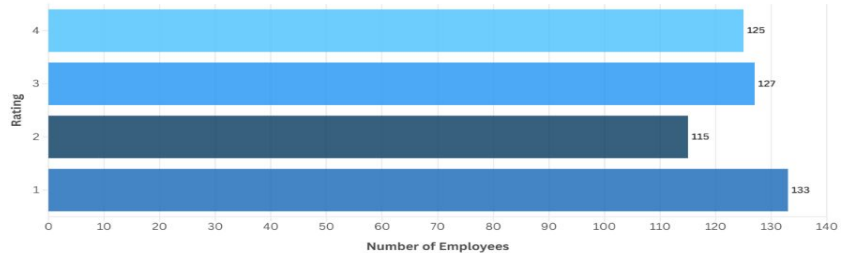
No Yes



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Attrition vs Job Satisfaction

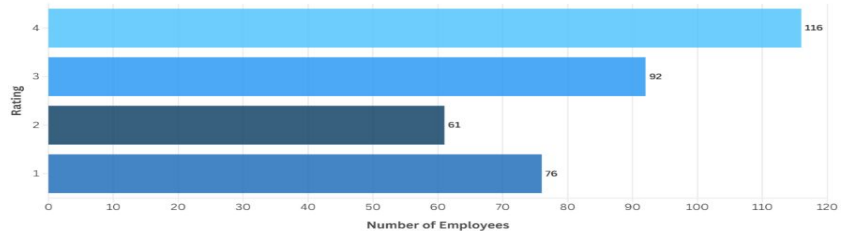
All No Yes



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Attrition vs Job Satisfaction

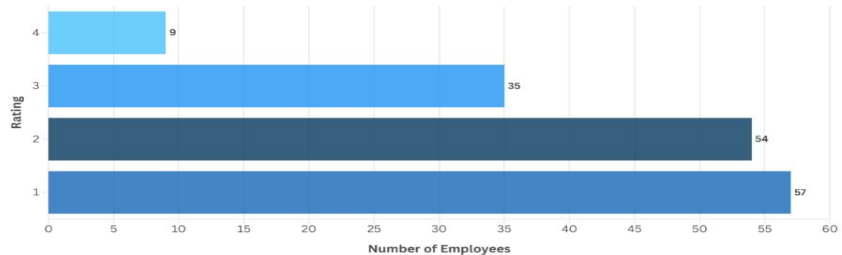
All No Yes



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Attrition vs Job Satisfaction

All No Yes



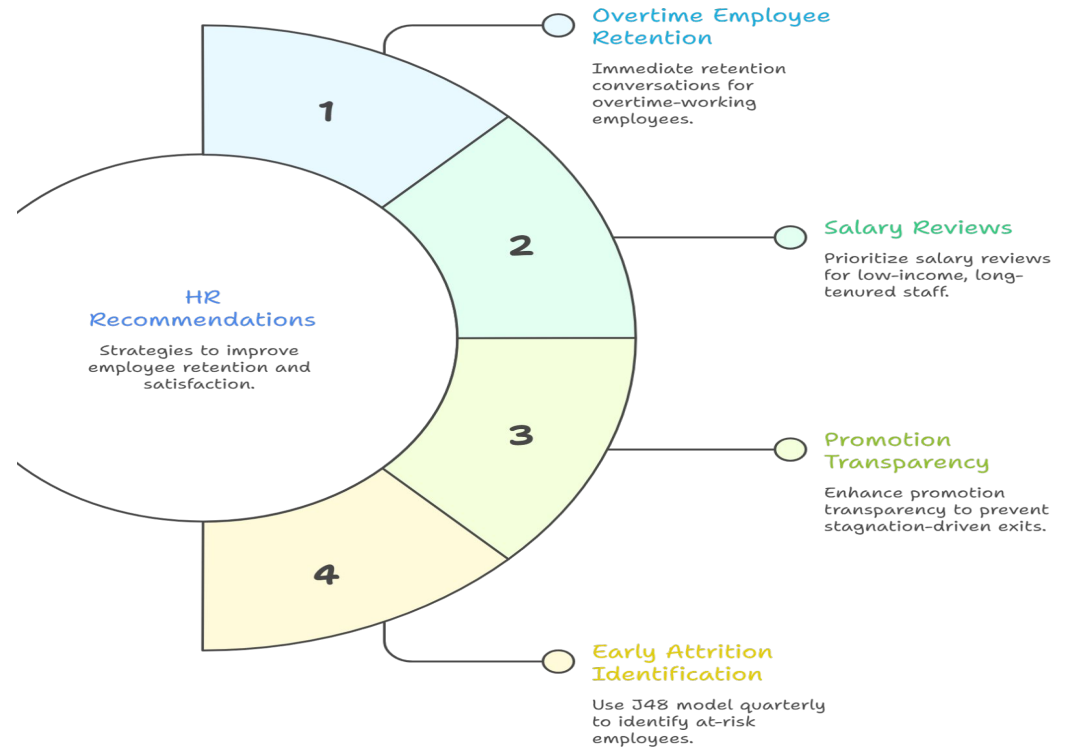
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Summary

Rising Employee Attrition at TechNova Solutions



Unveiling HR Strategies for Employee Retention



Reflection

1. Which attributes were most important for attrition?

The three most significant attributes were **OverTime**, **JobSatisfaction**, and **MonthlyIncome**. Employees working overtime were almost twice as likely to leave (45.7% attrition when OverTime = Yes vs. 24.6% when OverTime = No). Low JobSatisfaction scores strongly predicted departure — employees with satisfaction level 1 had 42.9% attrition and level 2 had 47.0% attrition, while level 4 had only 7.2% attrition. Lower MonthlyIncome was also a clear driver (employees who left earned on average ₹73,545 compared to ₹84,171 for those who stayed). YearsSinceLastPromotion showed a mild effect (average 3.19 years for those who left vs. 2.76 years for those who stayed). These findings directly support the corporate story's emphasis on work-life balance, growth opportunities, and fair compensation at TechNova Solutions.

2. Was your model overfitting or underfitting? Why?

Three models were trained to explore this. The **baseline pruned J48 decision tree** achieved approximately 49.9% accuracy and generalized reasonably well, making it the most reliable model. The **unpruned J48 (minNumObj=1)** demonstrated clear **overfitting** — training-set accuracy approached 98–100% but cross-validation accuracy dropped significantly, meaning the model memorized the training data rather than learning generalizable patterns. The **ZeroR** model demonstrated **underfitting** at approximately 69% accuracy — it was too simple and only predicted the majority “No” class without considering any features.

3. What data improvements could reduce errors?

The dataset has moderate class imbalance (31% Yes attrition). Applying SMOTE oversampling or cost-sensitive learning in Weka could help balance the classes. Adding more years of historical data (time-series) or qualitative fields such as manager feedback and exit-interview sentiment would make predictions more dynamic. Finally, collecting data across multiple years instead of a single snapshot would allow the model to detect early warning patterns before attrition actually occurs.

4. How should HR teams use this model in decision-making?

The baseline J48 model should be run quarterly on current employee data to generate an attrition risk score for each employee. HR teams can use this to prioritize retention interventions — employees flagged as high-risk should receive focused discussions on overtime reduction, salary reviews, and promotion planning. The model should be treated as an **early-warning tool**, not a replacement for human judgment. Managers and HR should use the predictions to start proactive conversations, not to label employees. Over time, as the company collects more real data, the model can be retrained to improve accuracy and adapt to changing organizational conditions.